

P100-067: Dataset on settling dynamics of microplastic particles sinking through exopolymer-induced rheological stratification

Principia-100 | Standardized portfolio | 1 October 2026

Scenario and evaluation contract

Domain: Environmental fluid mechanics. The current default task predicts Interface transit time in s.
Independent unit: complete fluid configuration; particles balanced within configuration. Its packaged cohort contains 76 assigned rows in 3 groups.

All supplied outcomes are now exposed. Historical reserved evaluations remain part of the evidence record; future scores are retrospective. No newly established fundamental law or measured deployment impact is admitted by this packaging.

Current portfolio findings

Unsupported or falsified: Dimensionless density/size and upstream-slowdown corrections fail robust transit transfer.

Limit: Development errors .528587 and .470794 absolute-log units do not establish a density/orientation mechanism; exposed simple ballistic control remains superior. Strict upstream windows and calibration timing limits are more solid than the rejected correction.

Unsupported or falsified: Blending near-entry and far-upstream velocities fails against the development kernel; later drag extensions remain unsupported.

Limit: Five of six configurations lose toRBF. The first continuation density/orientation corrections also failed exposed transfer.

Versioned tasks

Task family	Target and information	Source scope
original	Interface transit time (s)	original corpus
round2	Interface transit time (s)	original corpus

Different targets, calibration budgets and cohorts have different task IDs. Their raw error values cannot be pooled into a scenario ranking. Exact equations, coefficients, permitted variables, timing, groups and per-model results are bound in the task packages.

Evaluation: the shared CLI provides numerical scoring, explicit coverage and abstention, uncertainty and event diagnostics. Agent review separately assesses scientific support; no aggregate discovery score is produced.

Detailed reference note: Microplastic transit through a fluid interface

P100-067 | Scenario and findings | September 2026 reference results

The practical question is whether upstream particle motion predicts passage time through a visible rheological interface. In the reserved experiments, the elementary constant-speed transit approximation outperforms the proposed orientation correction. This is a useful scoped reproduction and a negative mechanistic finding.

1. Scenario and available measurements

The Mrokowska, Dzień and Krztoń-Maziopa dataset provides particle trajectories and orientation, particle descriptors, fluid rheology, density and refractive-index profiles across nine fluid configurations. The configurations belong to three numbered families. Multiple particle categories, including disks, rods and a sphere, allow geometry-dependent hypotheses to be tested alongside simpler transport descriptions.

The pilot's response is complete transit time across the visible interface, in seconds, rather than a full predicted trajectory. Interface width h is supplied by the recorded interface bounds. Upstream speed and orientation are obtained from observations strictly before entry. Source members, density-profile members and observation cutoffs remain attached to each response.

2. Experimental method

A fixed metadata-based allocation reserved one complete configuration per family: 1.1, 2.1 and 3.1. The remaining six configurations supported development. Five substantive attempts tested buoyancy, speed-gradient persistence, orientation-dependent drag, spatial adaptation and orientation fluctuations. Selection used complete development configurations, with all candidates and controls frozen before confirmation.

The reserved cohort contains 76 trajectories: 28, 23 and 25 in the three configurations. Linked representations are not independent replicates. Earlier analyses that interpolated predictor information across interface entry were invalidated; the retained evidence uses strict native upstream observations. Ambiguous thickness and fluid-filename metadata are not used to support thickness-dependent or normal-stress claims.

3. Equations and the proposed mechanism

The elementary transport comparator assumes that upstream speed persists over the interface width:

$$T_0 = \frac{h}{v_{\text{up}}}.$$

Here h is in metres and upstream speed is in m/s, giving T in seconds. The upstream speed is measured in a separate observation window, not calculated from the target transit, so this comparison is not an accounting identity.

For the tested orientation extension, define a from upstream orientation angle θ and disk/rod indicators I :

$$a = \frac{1 + \langle \cos(2\theta) \rangle}{2}, \quad \hat{T} = T_0 \exp(\eta).$$

$$\eta = b_0 + b_v \ln\left(\frac{v_{\text{up}}}{0.01 \text{ m s}^{-1}}\right) + b_D I_D a + b_R I_R a.$$

The dimensionless coefficients are (-0.371975, 0.211069, 0.350457, 1.051199), in the displayed order. The correction tests whether projected orientation acts as a transferable drag-state indicator. The stored orientation variable is mean $\cos(2\theta)$, not mean $\cos^2\theta$; the transformation above gives mean $\cos^2\theta$ for the same observations.

4. Findings and predictive performance

The elementary transit predictor wins all three reserved configurations. It has mean absolute log error 0.117418 and balanced physical MAE 0.047682 s. The selected orientation extension reaches 0.366041 and 0.230554 s, respectively. Thus, additional mechanistic-looking terms do not improve this transfer test.

Frozen model	Absolute log error	Balanced MAE (s)
Orientation extension, selected before confirmation	0.366041	0.230554
Constant upstream-speed comparator	0.117418	0.047682
Particle/family categorical comparator	0.359262	0.115559
Development mean-orientation control	0.266464	0.104537

For configuration g , particle category p and trajectory i , define absolute log error ℓ . Let K_g be the number of observed particle categories and n_{gp} their trajectory counts. The primary metric is:

$$\ell_{gpi} = \ln(\hat{T}_{gpi}/T_{gpi}), \quad E_{\log} = \frac{1}{3} \sum_{g=1}^3 \frac{1}{K_g} \sum_{p=1}^{K_g} \frac{1}{n_{gp}} \sum_{i=1}^{n_{gp}} \ell_{gpi}.$$

Particle categories receive equal weight within each configuration; configurations then receive equal weight. Physical MAE uses the same nesting. Log error is dimensionless, not a percentage accuracy. For the simple comparator, configuration-specific physical MAEs are 0.0260, 0.0241 and 0.0929 s, making the weaker transfer to configuration 3.1 visible.

Individual upstream orientation is not established as the transferable mechanism. The fitted orientation model also loses to its categorical comparator and to a control that substitutes development-family mean orientation. Together with the preserved development controls, this undermines the interpretation that individual orientation supplies robust predictive drag information in this experiment.

5. Practical value and limits

For a process with known interface bounds and observable upstream motion, the simple transit scale provides a transparent benchmark for retention-time or separation studies. The data indicate when a richer drag correction must justify itself against a parameter-free reference. Neither separator performance nor environmental residence time outside this apparatus was tested.

Constant-speed prediction succeeding here does not prove that drag is unchanged across the interface. Compensating accelerations, the observed window and limited configurations can all affect total transit time. The orientation proxy is two-dimensional; normal stresses are unmeasured. Three reserved configurations do not establish universal behavior across fluids, particle sizes or interfaces, and do not support a precise population confidence interval.

6. Evidence and use

Reproduction: `python run.py` replays the four models. `rules.json` gives exact coefficients and the development-only mean-orientation lookup. Native sample anchors and cutoffs are retained in `data/observations.csv.gz`; group results are in `evidence/`.

Evaluation: `evaluator/README.md` specifies positive transit predictions, exact sample alignment, nested grouping, abstention and separate scientific review. A valid alternative mechanism need not match the reference formula.

Status: the supported result is a scoped reproduction of the elementary transit approximation, not a new transport law. All confirmation targets are exposed; a revised mechanism needs fresh configurations for independent confirmation.

Source: Mrokowska, Dzień and Krztoń-Maziopa, [microplastic sinking experiments](#), Mendeley Data v1, DOI 10.17632/ccm9k8pjft.1. Exact source assets and hashes are in `rules.json`.